ANALYSTLAB AFRICA

Experience Lab Internship Programme

Official Weekly Assignment

Week 6: HealthConnect Rwanda Healthcare Analytics & Artificial Intelligence Transformation Project

Document Control

Programme	AnalystLab Africa Experience Lab
Week	Week 6
Project	HealthConnect Rwanda Healthcare Analytics & Artificial Intelligence Transformation Project
Project Title	Enterprise Healthcare Analytics, Artificial Intelligence and Digital Transformation System
Version	1.0
Issue Date	_____
Submission Deadline	End of Week 6 (Sunday, 11:59 PM WAT)
Submission Method	Official AnalystLab Africa Week 6 Submission Form

Project Overview

Healthcare organizations generate enormous volumes of clinical, operational, financial, and administrative data every day. These datasets contain valuable insights that can improve patient care, optimize healthcare operations, enhance financial performance, support disease surveillance, and enable evidence-based decision-making.

In this project, interns will work with a relational Healthcare Analytics Database consisting of **27** interconnected enterprise datasets covering healthcare facilities, medical staff, patients, appointments, diagnoses, laboratory tests, prescriptions, admissions, billing, insurance claims, payments, and patient feedback. Learners will perform data cleaning, exploratory data analysis, SQL querying, dashboard development, predictive modeling, Generative AI solution design, and project planning based on their respective learning tracks.

The objective of this project is to provide data-driven recommendations that improve healthcare service delivery, optimize hospital operations, enhance patient outcomes, support financial sustainability, and accelerate the digital transformation of HealthConnect Rwanda through analytics, Artificial Intelligence, and modern project management practices.

Learning Outcomes

By the end of this project, interns should be able to:

- Design and manage relational healthcare databases.
- Perform healthcare data cleaning, validation, and integration.
- Write advanced SQL queries to analyze clinical, operational, and financial data.
- Build interactive Power BI dashboards for healthcare executives.
- Analyze patient care, hospital operations, and healthcare performance using statistical techniques.
- Develop machine learning models for clinical risk prediction, healthcare forecasting, and operational optimization.
- Design AI-powered healthcare assistants and Retrieval-Augmented Generation (RAG) solutions using enterprise healthcare data.
- Apply project management methodologies to plan and manage healthcare analytics and AI implementation projects.
- Communicate actionable insights and strategic recommendations to healthcare executives and stakeholders.
- Demonstrate ethical, responsible, and secure use of data and Artificial Intelligence in healthcare environments.

Project Datasets

Interns will use the following enterprise healthcare datasets.

Dataset	Description
Countries	Country master information
Provinces	Province information
Districts	District information
Sectors	Sector information
Villages	Village information
Ownership Types	Government, Private, Faith-Based, NGO, Military, etc.
Health Facilities	Healthcare facility master information
Departments	Clinical and administrative departments
Medical Specialties	Doctor specialization information

Dataset	Description
Healthcare Services	Medical and healthcare services offered
Medical Equipment	Medical equipment inventory
Rooms	Hospital rooms and wards
Beds	Bed inventory and allocation
Doctors	Doctor master information
Nurses	Nurse master information
Pharmacists	Pharmacist master information
Laboratory Scientists	Laboratory scientist information
Radiographers	Radiographer information
Administrative Staff	Hospital administrative staff
Patients	Patient master records
Diagnoses	Patient diagnosis records
Clinical Transactions	Contains transactional datasets that capture the complete clinical journey of patients, including Appointments, Diagnoses, Laboratory Tests, Prescriptions, Admissions, and Referrals. These datasets record patient interactions with healthcare providers and support analysis of clinical workflows, disease trends, treatment outcomes, healthcare utilization, and operational performance.
Billing	Billing and invoicing records
Insurance Claims	Insurance claim transactions
Payments	Patient payment transactions
Patient Feedback	Patient satisfaction and feedback records

Dataset	Description
Revenue Summary	Monthly and facility revenue summary

Track 1 – Data Analytics

Project Title

Healthcare Operations and Performance Analytics Dashboard

Focus

Using SQL, Excel, and Power BI to analyze hospital operations, patient care, clinical services, and financial performance through executive dashboards and KPI reporting.

Track Overview

As a Data Analyst at HealthConnect Rwanda, you have been assigned to support executive management by transforming healthcare data into meaningful insights. Your responsibility is to analyze patient care, hospital operations, financial performance, and service delivery using SQL, Excel, and Power BI.

The executive board expects an interactive dashboard that enables management to monitor Key Performance Indicators (KPIs), identify operational bottlenecks, and make informed strategic decisions.

Learning Outcomes

At the end of this track, interns should be able to:

- Understand healthcare data models.
- Design healthcare KPIs.
- Perform exploratory data analysis.
- Develop executive dashboards.
- Build healthcare performance reports.
- Apply SQL for healthcare analytics.
- Communicate business insights effectively.

Business Objectives

Your analysis should help HealthConnect Rwanda answer the following business questions:

- Which hospitals serve the highest number of patients?

- Which departments experience the greatest workload?
- Which diseases are becoming more common?
- Are medical staff distributed efficiently?
- Which insurance companies generate the largest revenue?
- Which facilities have the highest patient satisfaction?
- Which hospitals require expansion?
- How can operational costs be reduced?
- Which services contribute most to revenue?
- Which regions require additional healthcare investment?

Key Performance Indicators (KPIs)

Develop KPIs for each of the following categories.

Patient KPIs

Calculate:

- Total Registered Patients
- Active Patients
- New Patients
- Returning Patients
- Male vs Female Patients
- Age Group Distribution
- Blood Group Distribution
- Insurance Coverage Rate
- Patients by Province
- Patients by District
- Patients by Chronic Disease
- Average Patient Age

Appointment KPIs

Calculate:

- Total Appointments
- Daily Appointments
- Monthly Appointments
- Appointment Growth Rate
- Emergency Appointments
- Follow-up Appointments
- Consultation Appointments
- Referral Rate
- Appointment Cancellation Rate (if applicable)
- Average Appointments per Doctor

Clinical KPIs

Develop dashboards showing:

- Top 20 Diagnoses
- Disease Severity Distribution
- Laboratory Test Volume
- Prescription Volume
- Admission Rate
- Average Length of Stay
- Referral Distribution
- Readmission Indicator
- Most Common Medications

Healthcare Staff KPIs

Calculate:

- Number of Doctors
- Number of Nurses
- Number of Pharmacists
- Number of Laboratory Scientists
- Number of Radiographers
- Staff Distribution by Facility
- Staff Distribution by Department
- Average Years of Experience
- Staff-to-Patient Ratio

Operational KPIs

Analyze:

- Number of Health Facilities
- Facility Type Distribution
- Bed Capacity
- Available Beds
- Bed Utilization Rate
- Room Utilization
- Equipment Distribution
- Equipment Availability
- Department Capacity
- Facility Performance Ranking

Financial KPIs

Develop dashboards showing:

- Total Revenue
- Monthly Revenue
- Revenue by Hospital
- Revenue by Department
- Revenue by Service
- Revenue by Province
- Revenue by Insurance Provider
- Outstanding Bills
- Insurance Claim Approval Rate
- Payment Method Distribution
- Average Billing Amount
- Revenue Growth Rate

Patient Experience KPIs

Analyze:

- Average Satisfaction Rating
- Rating Distribution
- Satisfaction by Hospital
- Satisfaction by Department
- Most Common Feedback
- Poor Rating Trend
- Excellent Rating Trend

SQL Tasks

Write SQL queries to answer the following business questions.

Patient Analysis

1. Retrieve all registered patients.
2. Count patients by gender.
3. Count patients by province.
4. Identify patients above 65 years.
5. Find patients with chronic diseases.
6. Retrieve patients covered by insurance.
7. Count patients by blood group.
8. List the youngest and oldest patients.
9. Identify duplicate phone numbers.
10. Calculate average patient age.

Appointment Analysis

11. Count appointments by month.
12. Identify busiest hospitals.
13. Find busiest departments.
14. Find doctors with the highest workload.
15. Identify referral patterns.
16. Count emergency appointments.
17. Calculate average appointments per doctor.
18. Retrieve appointments by province.
19. Identify facilities with the highest patient volume.
20. Find peak consultation periods.

Clinical Analysis

21. Top diagnoses.
22. Most requested laboratory tests.
23. Most prescribed medications.
24. Average admission duration.
25. Disease severity distribution.
26. Laboratory test utilization.
27. Prescription trends.
28. Referral destinations.
29. Admission rate by diagnosis.
30. Clinical workload by department.

Financial Analysis

31. Revenue by facility.
32. Revenue by department.
33. Revenue by service.
34. Insurance claims by provider.
35. Outstanding payments.
36. Average billing amount.
37. Monthly revenue trend.
38. Payment methods.
39. Top revenue-generating hospitals.
40. Financial performance ranking.

Power BI Dashboard

Create a professional executive dashboard containing:

Page 1 – Executive Dashboard

Include:

- Total Patients
- Total Appointments
- Total Revenue
- Total Facilities
- Total Doctors
- Average Patient Satisfaction

Page 2 – Patient Analytics

Visualizations:

- Gender Distribution
- Age Pyramid
- Province Map
- Chronic Disease Analysis
- Insurance Distribution

Page 3 – Clinical Analytics

Visualizations:

- Diagnosis Analysis
- Laboratory Tests
- Prescriptions
- Admissions
- Referrals

Page 4 – Financial Dashboard

Include:

- Revenue Trend
- Revenue by Facility
- Revenue by Department
- Insurance Claims
- Payment Methods

Page 5 – Operational Dashboard

Include:

- Staff Distribution

- Bed Utilization
- Equipment Utilization
- Facility Comparison
- Department Comparison

Page 6 – Patient Experience Dashboard

Include:

- Satisfaction Ratings
- Feedback Analysis
- Complaint Trends
- Hospital Ranking
- Department Ranking

Required Power BI Features

Your dashboard must demonstrate the use of:

- KPI Cards
- Slicers
- Drill-through Pages
- Bookmarks
- Tooltips
- Hierarchies
- Conditional Formatting
- Maps
- Matrix Tables
- DAX Measures
- Time Intelligence
- Dynamic Titles
- Decomposition Tree
- Key Influencers
- Forecasting (where applicable)

Deliverables

Each intern should submit:

- SQL Script (.sql)
- Power BI Dashboard (.pbix)
- Executive Dashboard PDF
- Business Insights Report (3–5 pages)
- Executive Presentation (10–15 slides)

ASSESSMENT CRITERIA

Section	Marks
Data Cleaning, Integration & Validation	15
SQL Queries & Healthcare Data Analysis	20
Power BI Dashboard Design (KPIs, DAX, Visualizations)	25
Business Insights & Executive Recommendations	15
Dashboard Usability, Interactivity & Storytelling	10
Documentation & Presentation	15
Total	**100 Marks**

Submission Requirements

Interns should submit:

- PostgreSQL Database Backup (.sql or .backup)
- SQL Query Scripts
- Power BI Dashboard (.pbix)
- Business Insights Report (PDF/Word)
- README.md
- GitHub Repository Link (if applicable)
- Presentation Slides (PDF/PPT)

Expected Deliverables

- Fully implemented PostgreSQL database
- Cleaned and validated datasets
- SQL scripts answering the required business questions
- Interactive Power BI dashboard
- Business insights and recommendations report
- Project documentation

2. DATA SCIENCE TRACK

Project Title

Healthcare Predictive Analytics and Population Health Intelligence

Focus

Applying statistical analysis, exploratory data analysis (EDA), feature engineering, forecasting, and predictive analytics to identify healthcare trends and support evidence-based decision-making.

Track Overview

As a **Data Scientist** at HealthConnect Rwanda, you have been assigned to uncover hidden patterns in healthcare data, identify key drivers of patient outcomes, predict operational trends, and support executive decision-making using statistical and predictive analytics.

You will work with enterprise healthcare datasets containing patient demographics, clinical records, healthcare operations, financial transactions, and patient feedback to generate actionable insights.

Learning Outcomes

At the end of this track, interns should be able to:

- Understand enterprise healthcare datasets.
- Perform data preprocessing and cleaning.
- Conduct exploratory and statistical data analysis.
- Engineer predictive features.
- Identify trends and patterns using data science techniques.
- Build healthcare forecasting models.
- Present data-driven recommendations.

Business Objectives

HealthConnect Rwanda would like to answer the following strategic questions:

- Which factors influence hospital admissions?
- Which patient groups are most at risk of chronic diseases?
- What factors affect patient satisfaction?
- Can future patient demand be forecast accurately?
- Which departments are likely to become overloaded?
- How can hospital resources be allocated more efficiently?
- What operational improvements will maximize healthcare outcomes?

Data Preparation Tasks

Using Python (Pandas) or R (tidyverse):

Task 1 – Data Loading

- Import all enterprise datasets.
- Merge related tables using appropriate keys.
- Verify relationships between datasets.

Task 2 – Data Cleaning

Perform:

- Missing value detection
- Duplicate removal
- Invalid record detection
- Data type conversion
- Date formatting
- Category standardization

Task 3 – Data Validation

Validate:

- Foreign key relationships
- Duplicate Patient IDs
- Duplicate Appointment IDs
- Invalid Facility IDs
- Invalid Department IDs
- Impossible Ages
- Invalid Billing Amounts

Prepare a **Data Quality Report** summarizing all findings.

Exploratory Data Analysis (EDA)

Patient Analysis

Analyze:

- Gender distribution
- Age distribution
- Blood group distribution
- Insurance coverage
- Chronic disease prevalence
- Geographic distribution
- Patient growth trend

Clinical Analysis

Analyze:

- Diagnosis frequency
- Disease severity
- Laboratory test utilization
- Prescription frequency
- Admission trend
- Referral trend

Operational Analysis

Analyze:

- Doctor workload
- Nurse workload
- Facility utilization
- Bed occupancy
- Equipment distribution
- Staff-to-patient ratio

Financial Analysis

Analyze:

- Revenue trend
- Billing distribution
- Insurance claims
- Outstanding payments
- Payment methods
- Revenue by department

Patient Experience

Analyze:

- Satisfaction scores
- Feedback trends
- Department comparison
- Facility comparison

Feature Engineering

Create new variables including:

- Patient Age Group
- High-Risk Patient Indicator
- Length of Stay Category
- Frequent Visitor Indicator
- Insurance Coverage Category
- Revenue Category
- Disease Severity Score
- Facility Performance Score
- Patient Satisfaction Category
- Department Workload Index

Statistical Analysis

Conduct the following analyses:

Descriptive Statistics

Calculate:

- Mean
- Median
- Mode
- Standard Deviation
- Variance
- Quartiles
- Percentiles

Correlation Analysis

Investigate relationships between:

- Age and Hospital Admissions
- Age and Billing Amount
- Diagnosis and Length of Stay
- Department Workload and Satisfaction
- Revenue and Patient Volume
- Insurance Coverage and Healthcare Utilization

Visualize findings using a correlation heatmap.

Hypothesis Testing

Perform and interpret at least three statistical tests, for example:

1. Compare average billing amounts between insured and uninsured patients.
2. Determine whether patient satisfaction differs significantly across facilities.
3. Test whether admission rates differ between age groups.

Clearly state the null hypothesis, alternative hypothesis, test used, significance level, p-value, and conclusion.

Predictive Analytics

Develop predictive models to answer the following business questions:

Task 1 – Patient Volume Forecasting

Forecast monthly patient visits for the next 12 months using time-series techniques.

Task 2 – Revenue Forecasting

Predict monthly hospital revenue for the next 12 months.

Task 3 – Hospital Admission Prediction

Estimate the probability that a patient will require admission based on demographics, diagnosis, and appointment details.

Task 4 – Patient Satisfaction Prediction

Predict patient satisfaction ratings using operational and clinical variables.

Task 5 – Department Workload Forecast

Forecast future patient demand for each department to support staffing and resource planning.

Clustering Analysis

Use clustering techniques (e.g., K-Means or Hierarchical Clustering) to:

- Segment patients into meaningful groups.
- Identify high-risk patient clusters.
- Group facilities based on operational performance.
- Segment departments by workload and service utilization.

Interpret the clusters and discuss how management can use them.

Dimensionality Reduction (Optional)

Apply PCA (Principal Component Analysis) to reduce feature dimensions and explain the proportion of variance retained.

Data Visualization

Create high-quality visualizations such as:

- Histograms
- Box Plots
- Violin Plots
- Heatmaps
- Pair Plots
- Correlation Matrix
- Time-Series Line Charts
- Scatter Plots
- Density Plots
- Geographic Maps

Ensure every chart includes an appropriate title, axis labels, and a brief interpretation.

Business Recommendations

Prepare a report addressing:

- Key operational challenges identified.
- High-risk patient groups.
- Opportunities to improve patient satisfaction.
- Resource optimization strategies.
- Revenue improvement opportunities.
- Recommendations for expanding healthcare services.

Support every recommendation with evidence from your analysis.

Deliverables

Each intern should submit:

- Python Notebook (.ipynb) or R Notebook (.Rmd)
- Cleaned and integrated dataset
- Data Quality Report
- Exploratory Data Analysis Report
- Statistical Analysis Report
- Predictive Analytics Report
- Visualizations
- Executive Summary (3–5 pages)

- Presentation Slides (10–15 slides)

ASSESSMENT CRITERIA

Section	Marks
Data Preparation & Feature Engineering	20
Exploratory Data Analysis & Statistical Analysis	20
Predictive Analytics & Forecasting	20
Data Visualization & Interpretation	15
Business Recommendations & Healthcare Insights	10
Documentation & Presentation	15
Total	**100 Marks**

Machine Learning Track

Project Title

AI-Powered Clinical Risk Prediction and Healthcare Intelligence System

Focus

Developing machine learning models to predict hospital admissions, patient risk, insurance claim outcomes, patient satisfaction, and healthcare resource utilization.

Track Overview

As a **Machine Learning Engineer** at HealthConnect Rwanda, your responsibility is to develop predictive models that improve patient care, optimize hospital operations, reduce operational costs, and support executive decision-making.

Using enterprise healthcare data, you will design, train, evaluate, and interpret machine learning models capable of predicting healthcare outcomes and assisting clinicians and hospital administrators.

Your solutions should not only achieve good predictive performance but also provide interpretable insights that support responsible decision-making in a healthcare environment.

Learning Outcomes

At the end of this track, interns should be able to:

- Understand the end-to-end machine learning lifecycle.
- Prepare healthcare data for predictive modelling.
- Engineer features suitable for machine learning algorithms.
- Build classification, regression, clustering, and anomaly detection models.
- Evaluate and compare model performance.
- Interpret model predictions using Explainable AI (XAI) techniques.
- Communicate machine learning findings to technical and non-technical stakeholders.

Business Objectives

HealthConnect Rwanda would like to use Machine Learning to answer the following business questions:

- Which patients are likely to require hospital admission?
- Which appointments are likely to result in referrals?
- Which patients are at high risk of chronic disease progression?
- Which insurance claims are likely to be rejected?
- Which patients are likely to provide poor satisfaction ratings?
- Which facilities are likely to experience overcrowding?
- Which departments will require additional staffing?
- How can healthcare resources be allocated more efficiently?

Part A: Data Preparation

Task 1 – Data Integration

Merge relevant datasets into a machine-learning-ready dataset.

Recommended tables include:

- Patients
- Appointments
- Diagnoses
- Laboratory Tests
- Prescriptions
- Admissions
- Billing
- Insurance Claims
- Healthcare Staff
- Health Facilities

Document the relationships between the datasets.

Task 2 – Data Cleaning

Perform:

- Missing value treatment
- Duplicate removal
- Outlier detection
- Invalid record correction
- Encoding of categorical variables
- Scaling and normalization of numerical variables

Provide a summary of all preprocessing steps.

Task 3 – Feature Engineering

Create meaningful predictive features such as:

- Patient Age Group
- Number of Previous Appointments
- Number of Admissions
- Number of Chronic Conditions
- Previous Laboratory Tests
- Previous Prescriptions
- Insurance Coverage Status
- Distance to Health Facility (if geographic data is used)
- Department Workload
- Doctor Workload
- Facility Utilization Rate
- Average Waiting Time (derived if possible)
- Previous Satisfaction Score
- Revenue Category
- Seasonal Visit Indicator

Explain why each engineered feature is important.

Part B: Classification Models

Develop classification models for the following business problems.

Model 1 – Hospital Admission Prediction

Objective

Predict whether a patient will be admitted after an appointment.

Possible algorithms:

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost
- LightGBM
- CatBoost

Evaluate using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- Confusion Matrix

Model 2 – Patient Satisfaction Prediction

Predict whether a patient will provide:

- Low Satisfaction
- Medium Satisfaction
- High Satisfaction

Discuss factors that influence patient satisfaction.

Model 3 – Insurance Claim Approval Prediction

Predict whether an insurance claim will be:

- Approved
- Pending
- Rejected

Identify the most influential features.

Model 4 – High-Risk Patient Classification

Predict whether a patient is at **High Risk** or **Low Risk** based on:

- Age
- Chronic conditions
- Previous admissions
- Diagnosis
- Laboratory results

- Medication history

Recommend appropriate interventions for high-risk patients.

Part C: Regression Models

Develop regression models for the following tasks.

Model 5 – Revenue Prediction

Predict billing amount for each appointment.

Possible algorithms:

- Linear Regression
- Random Forest Regressor
- XGBoost Regressor
- Gradient Boosting Regressor

Evaluate using:

- RMSE
- MAE
- R^2 Score
- MAPE

Model 6 – Length of Stay Prediction

Predict the expected hospital stay (in days) for admitted patients.

Discuss how this prediction could improve bed management and resource allocation.

Part D: Clustering

Use unsupervised learning techniques to segment the data.

Patient Segmentation

Cluster patients based on:

- Age

- Chronic diseases
- Frequency of visits
- Insurance type
- Billing history
- Diagnosis patterns

Interpret each cluster and recommend tailored healthcare strategies.

Facility Segmentation

Cluster healthcare facilities based on:

- Patient volume
- Revenue
- Bed occupancy
- Staff numbers
- Satisfaction ratings

Explain how the clusters could support strategic planning.

Part E: Anomaly Detection

Develop an anomaly detection solution to identify:

- Unusually high billing amounts.
- Suspicious insurance claims.
- Abnormal laboratory results.
- Duplicate patient records.
- Unusual prescribing patterns.
- Facilities with unexpectedly low or high patient volumes.

Discuss how anomaly detection can reduce fraud and improve data quality.

Part F: Model Explainability (Explainable AI)

Interpret the best-performing model using Explainable AI techniques such as:

- Feature Importance
- SHAP (SHapley Additive exPlanations)
- LIME (Local Interpretable Model-agnostic Explanations)
- Partial Dependence Plots (optional)

Explain how model transparency supports trust in healthcare decision-making.

Part G: Model Evaluation and Comparison

Compare all developed models using a summary table that includes:

- Algorithm
- Business Problem
- Evaluation Metrics
- Strengths
- Limitations
- Recommended Use Case

Justify the final model selected for deployment.

Part H: Model Deployment (Conceptual)

Design a high-level architecture for deploying the selected model in a hospital environment.

Your design should describe:

- Data sources.
- Data preprocessing pipeline.
- Model serving layer.
- API integration with the Hospital Information System (HIS).
- Dashboard integration.
- Monitoring and retraining strategy.

A simple architecture diagram is encouraged.

Ethical and Responsible AI

Discuss the ethical considerations of using machine learning in healthcare, including:

- Patient privacy and confidentiality.
- Data security.
- Bias and fairness.
- Explainability.
- Human oversight.
- Responsible AI governance.

Deliverables

Each intern should submit:

1. Machine Learning Notebook (.ipynb)
2. Cleaned and feature-engineered dataset
3. Model Evaluation Report
4. Explainable AI Report
5. Confusion Matrices and Evaluation Metrics
6. Model Comparison Table
7. Deployment Architecture Diagram
8. Executive Summary (3–5 pages)
9. Presentation Slides (10–15 slides)

ASSESSMENT CRITERIA

Section	Marks
Data Preprocessing & Feature Engineering	15
Model Development (Classification, Regression, Clustering)	30
Model Evaluation & Explainable AI (SHAP/LIME)	20
Deployment Architecture & Business Application	10
Healthcare Recommendations & Ethical AI Considerations	10
Documentation & Presentation	15
Total	**100 Marks**

Generative AI Engineering Track

Project Title

Enterprise Healthcare AI Assistant and Clinical Decision Support System

Focus

Designing Generative AI solutions, Retrieval-Augmented Generation (RAG), AI agents, and intelligent assistants to automate clinical documentation, executive reporting, and healthcare knowledge retrieval.

Track Overview

As a **Generative AI Engineer** at **HealthConnect Rwanda**, you have been assigned to design intelligent AI-powered healthcare solutions that improve patient care, enhance hospital operations, support clinicians, and assist executive decision-making.

Using enterprise healthcare datasets, you will build AI systems capable of understanding natural language, retrieving relevant information, generating reports, summarizing clinical records, answering business questions, and supporting healthcare professionals while ensuring responsible and ethical AI practices.

Learning Outcomes

At the end of this track, interns should be able to:

- Understand Generative AI concepts and Large Language Models (LLMs).
- Design effective prompts for healthcare applications.
- Build Retrieval-Augmented Generation (RAG) systems.
- Develop AI-powered assistants using enterprise healthcare data.
- Integrate LLMs with structured datasets.
- Evaluate AI-generated outputs.
- Apply Responsible AI principles in healthcare.
- Communicate AI-generated insights effectively.

Business Scenario

HealthConnect Rwanda manages thousands of patients, appointments, diagnoses, prescriptions, laboratory results, insurance claims, billing transactions, and patient feedback records every month.

Hospital executives and clinicians spend significant time searching for information, writing reports, preparing summaries, and responding to operational questions.

Management has asked your team to develop Generative AI solutions that reduce administrative workload, improve access to information, and support evidence-based decision-making.

Business Objectives

Design AI solutions capable of:

- Summarizing patient medical histories.
- Explaining laboratory results in plain language.
- Generating discharge summaries.
- Answering executive questions about hospital performance.
- Producing monthly operational reports.
- Assisting clinicians during consultations.
- Supporting hospital administrators with strategic insights.
- Improving patient communication through AI.

Part A – Prompt Engineering

Design effective prompts for the following use cases.

Task 1 – Patient Summary Prompt

Develop a prompt that generates a concise patient summary including:

- Demographics
- Medical history
- Diagnosis
- Laboratory tests
- Medications
- Admission history
- Insurance information
- Follow-up recommendations

Task 2 – Executive Dashboard Prompt

Design a prompt that enables an executive to ask questions such as:

- Which hospitals generated the highest revenue this month?
- Which departments experienced the highest patient volume?
- Which diseases are increasing?
- Which facilities require additional staffing?

Task 3 – Clinical Decision Support Prompt

Create prompts that assist doctors in:

- Reviewing patient history.
- Identifying possible risk factors.
- Recommending additional investigations.
- Suggesting follow-up actions.

Note: Recommendations should support, not replace, clinical judgment.

Task 4 – Patient Communication Prompt

Design prompts that generate:

- Appointment reminders.
- Medication reminders.
- Laboratory test instructions.
- Discharge instructions.
- Health education messages.

Ensure the language is simple and patient-friendly.

Part B – Retrieval-Augmented Generation (RAG)

Design a RAG solution that retrieves information from the enterprise healthcare datasets before generating responses.

Your design should include:

- Data sources.
- Document indexing.
- Embedding generation.
- Vector database.
- Retrieval process.
- LLM response generation.

Create a simple architecture diagram illustrating the workflow.

Part C – AI Healthcare Assistant

Design an AI assistant capable of answering questions such as:

Patient Queries

- What medications was Patient X prescribed?
- When is the next appointment?
- Which laboratory tests were performed?
- What is the patient's diagnosis?

Executive Queries

- Show monthly revenue.
- Compare hospital performance.
- Identify the busiest departments.
- Which facilities have low patient satisfaction?
- Which insurance provider processes the highest number of claims?

Operational Queries

- Current bed availability.
- Doctor availability.
- Equipment utilization.
- Staff workload.
- Admission trends.

Document how the AI assistant retrieves and presents information.

Part D – AI Report Generation

Develop prompts that automatically generate:

Clinical Reports

- Patient Summary
- Discharge Summary
- Referral Letter
- Laboratory Summary
- Medication Summary

Executive Reports

Generate:

- Weekly Hospital Performance Report
- Monthly Revenue Report
- Executive KPI Dashboard Narrative
- Patient Satisfaction Report
- Healthcare Operations Report

Reports should be clear, concise, and suitable for executive audiences.

Part E – AI-Powered Insights

Using the enterprise datasets, design prompts that enable the AI to:

- Identify operational bottlenecks.
- Detect unusual trends.
- Explain changes in revenue.
- Recommend staffing improvements.
- Suggest capacity expansion.
- Identify high-risk patient groups.
- Highlight declining patient satisfaction.

Each insight should include supporting evidence from the data.

Part F – AI Agent Design

Design an intelligent Healthcare AI Agent with the following capabilities:

- Answer patient questions.
- Retrieve patient records.
- Explain billing information.
- Summarize medical history.
- Generate appointment reminders.
- Recommend appropriate hospital departments.
- Assist hospital administrators with reporting.

Provide a workflow diagram showing how users interact with the agent.

Part G – Responsible AI

Discuss how HealthConnect Rwanda should ensure responsible AI deployment by addressing:

- Patient privacy and confidentiality.
- Data security.
- Transparency and explainability.
- Bias and fairness.
- Human oversight.
- Regulatory compliance.
- AI governance.
- Risk management.

Recommend practical safeguards for each area.

Part H – AI Solution Evaluation

Evaluate your proposed AI solution using criteria such as:

- Accuracy
- Relevance
- Response quality
- Hallucination risk
- Retrieval quality
- Prompt effectiveness
- User experience
- Ethical compliance
- Scalability

Suggest methods for continuous monitoring and improvement.

Deliverables

Each intern should submit:

1. Prompt Engineering Document (minimum 20 well-designed prompts)
2. Retrieval-Augmented Generation (RAG) Architecture Diagram
3. AI Healthcare Assistant Design Document
4. AI Agent Workflow Diagram
5. AI-Generated Clinical Reports
6. AI-Generated Executive Reports
7. Responsible AI Assessment Report
8. Final Technical Report (5–8 pages)
9. Executive Presentation (10–15 slides)
10. Source code or prototype (if implemented)

ASSESSMENT CRITERIA

Section	**Marks**
Prompt Engineering	15
RAG Architecture & Knowledge Retrieval Design	20
AI Healthcare Assistant / AI Agent Design	20
AI-Generated Reports, Clinical Summaries & Executive Insights	20
Responsible AI, Security & Governance	10
Documentation & Presentation	15
Total	**100 Marks**

Project Management Track

Project Title

Healthcare Analytics and AI Transformation Programme Implementation Project

Focus

Planning, managing, and governing the implementation of an enterprise healthcare analytics and AI platform using project management best practices, stakeholder management, Agile delivery, risk management, and change management.

Track Overview

As the **Project Manager** for the **HealthConnect Rwanda Analytics & AI Transformation Project**, you have been appointed to lead the successful planning, execution, monitoring, and delivery of an enterprise-wide healthcare analytics initiative.

The project aims to integrate healthcare data from multiple systems, improve operational efficiency, enhance patient care, support executive decision-making, and introduce Artificial Intelligence into healthcare operations.

You are responsible for ensuring that the project is delivered on time, within budget, within scope, and meets stakeholder expectations while managing risks, communication, quality, and change.

Learning Outcomes

At the end of this track, interns should be able to:

- Develop professional project management documentation.
- Apply PMBOK principles throughout the project lifecycle.
- Manage healthcare stakeholders effectively.
- Develop project schedules and resource plans.
- Identify and mitigate project risks.
- Manage project communications.
- Monitor project performance using KPIs.
- Deliver executive project reports.

Business Scenario

HealthConnect Rwanda has approved a strategic initiative to implement an integrated Healthcare Analytics and Artificial Intelligence platform across its hospitals and healthcare facilities.

The project will integrate data from:

- Electronic Medical Records (EMR)
- Hospital Information Systems (HIS)
- Laboratory Information Systems (LIS)
- Pharmacy Management Systems
- Billing Systems
- Insurance Systems
- Patient Feedback Platforms

The Executive Board expects the implementation to improve patient care, optimize hospital operations, increase revenue, reduce costs, and support evidence-based decision-making.

Your consulting team has been contracted to plan and manage the successful delivery of this project.

Project Objectives

The project should achieve the following objectives:

- Integrate enterprise healthcare datasets into a unified analytics platform.
- Develop executive dashboards.
- Improve patient care through analytics and AI.

- Enhance financial reporting.
- Improve operational efficiency.
- Support data-driven decision-making.
- Develop AI-powered healthcare assistants.
- Build predictive healthcare models.

Part A – Project Initiation

Task 1 – Business Case

Prepare a Business Case covering:

- Business problem
- Current challenges
- Project justification
- Expected benefits
- Estimated costs
- Expected Return on Investment (ROI)
- Strategic alignment with organizational goals

Task 2 – Project Charter

Develop a professional Project Charter including:

- Project Title
- Business Need
- Project Objectives
- Scope
- Assumptions
- Constraints
- Deliverables
- Success Criteria
- Budget Estimate
- Project Sponsor
- Project Manager
- Approval Section

Task 3 – Stakeholder Register

Identify key stakeholders, including:

- Executive Board
- Ministry of Health
- Hospital Directors
- Doctors
- Nurses
- Pharmacists
- Laboratory Scientists
- Patients
- Insurance Providers

- IT Department
- Data Analysts
- Data Scientists
- Machine Learning Engineers
- Generative AI Engineers
- Vendors

For each stakeholder, document:

- Role
- Influence
- Interest
- Expectations
- Communication Needs

Part B – Project Planning

Task 4 – Scope Management

Develop:

- Project Scope Statement
- In-Scope Items
- Out-of-Scope Items
- Acceptance Criteria
- Scope Baseline

Task 5 – Work Breakdown Structure (WBS)

Develop a WBS that includes:

1. Project Initiation
2. Requirements Gathering
3. Data Collection
4. Data Cleaning
5. Data Integration
6. Dashboard Development
7. Machine Learning Development
8. Generative AI Development
9. Testing
10. Deployment
11. User Training
12. Project Closure

Task 6 – Project Schedule

Prepare a project schedule including:

- Major milestones
- Task durations

- Dependencies
- Critical Path (optional)
- Gantt Chart

Task 7 – Resource Plan

Prepare a resource allocation plan showing:

- Human Resources
- Hardware
- Software
- Cloud Services
- Budget Allocation
- Training Requirements

Task 8 – Budget Planning

Estimate project costs for:

- Personnel
- Infrastructure
- Software Licenses
- Cloud Services
- Training
- Data Storage
- AI Development
- Contingency Reserve

Provide a high-level project budget and justify key assumptions.

Part C – Risk Management

Task 9 – Risk Register

Identify at least **15 project risks**, including:

- Data quality issues
- Scope creep
- Budget overruns
- Delayed stakeholder approvals
- Cybersecurity threats
- Regulatory compliance
- Staff resistance to change
- AI model inaccuracies
- Vendor delays
- Infrastructure failures

- Data privacy breaches
- Resource shortages
- Schedule delays
- System integration challenges
- Low user adoption

For each risk, document:

- Probability
- Impact
- Risk Rating
- Mitigation Strategy
- Contingency Plan
- Risk Owner

Task 10 – Risk Matrix

Create a Probability–Impact Matrix and classify risks as:

- Low
- Medium
- High
- Critical

Prioritize the top five risks for immediate management attention.

Part D – Communication Management

Task 11 – Communication Plan

Develop a Communication Plan that identifies:

- Stakeholder Group
- Information Required
- Communication Method
- Frequency
- Responsible Person

Include communications for:

- Weekly project meetings
- Executive status reports
- Sprint reviews
- Risk updates
- Steering committee meetings
- User training sessions

Part E – Quality Management

Task 12 – Quality Management Plan

Develop quality standards for:

- Data Quality
- Dashboard Quality
- Machine Learning Models
- AI Solutions
- Documentation
- Testing
- User Acceptance Testing (UAT)

Define quality metrics and acceptance criteria.

Part F – Agile Project Management

Task 13 – Product Backlog

Develop a Product Backlog containing at least **25 user stories**.

Example:

As a Hospital Administrator, I want to view monthly revenue by facility so that I can identify underperforming hospitals.

Prioritize user stories using MoSCoW (Must, Should, Could, Won't).

Task 14 – Sprint Planning

Plan four two-week sprints, defining:

- Sprint Goal
- Selected User Stories
- Expected Deliverables
- Definition of Done
- Sprint Review Activities

Part G – Change Management

Task 15 – Change Management Plan

Develop a plan covering:

- Change Request Process
- Change Approval Workflow

- Impact Assessment
- Stakeholder Engagement
- Training Strategy
- Communication Strategy
- Adoption Metrics

Explain how you would encourage adoption of the new analytics and AI platform.

Part H – Monitoring and Control

Task 16 – Project Dashboard

Design a Project Dashboard that tracks:

- Schedule Performance
- Budget Performance
- Risk Status
- Issue Log
- Milestone Progress
- Resource Utilization
- Quality Metrics
- Stakeholder Satisfaction

Recommend suitable visualization tools (e.g., Microsoft Project, Power BI, Jira, Trello).

Part I – Project Closure

Task 17 – Project Closure Report

Prepare a Project Closure Report including:

- Project Summary
- Deliverables Completed
- Budget Performance
- Schedule Performance
- Lessons Learned
- Outstanding Issues
- Recommendations for Future Projects
- Formal Sign-off

Ethical, Legal, and Regulatory Considerations

Discuss how the project will ensure compliance with:

- Patient privacy and confidentiality.
- Data protection regulations.
- Information security policies.
- Ethical AI principles.

- Healthcare governance.
- Change management best practices.

Deliverables

Each intern should submit:

1. Business Case
2. Project Charter
3. Stakeholder Register
4. Scope Statement
5. Work Breakdown Structure (WBS)
6. Gantt Chart / Project Schedule
7. Resource Plan
8. Budget Estimate
9. Risk Register
10. Probability–Impact Matrix
11. Communication Plan
12. Quality Management Plan
13. Product Backlog
14. Sprint Plan
15. Change Management Plan
16. Project Dashboard
17. Project Closure Report
18. Executive Summary (3–5 pages)
19. Presentation Slides (10–15 slides)

ASSESSMENT CRITERIA

Section	**Marks**
Project Initiation	15
Project Planning	25
Risk, Quality, Communication & Change Management	15
Project Monitoring & Reporting	20
Project Closure & Business Value	10
Documentation & Presentation	15
Total	**100 Marks**

Professional Development Requirement

As part of building your professional portfolio:

• Update your GitHub repository with your Week 2 project.

• Publish a LinkedIn post describing your project

• Publish an X (Twitter) post summarizing your project.

• Tag AnalystLab Africa on LinkedIn.

• Tag @analystlabafric on X.

• Include the hashtag #AnalystLabAfrica on both platforms.

Professional Branding Reminder

AnalystLab Africa encourages every intern to document their learning journey professionally on LinkedIn and X. Consistently sharing your work helps build your portfolio and increases your visibility to recruiters and employers.